

Pre-screening

Participant	014
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	Yes	No
Do you or have you had a history of migraine headaches?	☐	☒
Do you or have you had a history of claustrophobia?	☐	☒
Do you or have you had a history of frequent or severe motion sickness?	☐	☒
Do you or have you had a history of any health problems (e.g., seizures, diabetes, heart problems, vertigo) that affect your ability to drive?	☐	☒
If you are a female, are you or is there a possibility that you might be pregnant?	☐	☒

Demographics

1. What is your race/ethnicity?	☒ White / Caucasian
2. What is your gender?	☒ Male
3. How old are you?	30
4. What is the highest level of education you have completed?	☒ Some college/Associates/Technical school training
5. Which option best describes your current home?	☒ It is owned or being bought by you or someone in household.
6. How do you pay for healthcare?	☒ Private insurance
8. Which best corresponds to your current work situation?	☒ Working full time.
9. What is your combined family income for the past 12 months (from all sources)	☒ \$50,000- \$99,999

Timeline Followback

Visit 1

Visit 1 Date

06-04-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 2

Visit 2 Date

07-15-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 3

Visit 3 Date

09-09-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Sunday Monday Tuesday Wednesday Thursday Friday
Saturday

on prescribed pain medications

____ e-Cigarettes or
nicotine vape, Other tobacco, Buprenorphine, Other
opioids e-Cigarettes or nicotine vape, Other tobacco,
Marijuana, Buprenorphine, Other opioids e-Cigarettes
or nicotine vape, Other tobacco, Buprenorphine, Other
opioids

Sunday Monday Tuesday Wednesday Thursday Friday
Saturday

e-Cigarettes or nicotine vape, Other tobacco,
Marijuana, Other opioids e-Cigarettes or nicotine
vape, Other tobacco, Other opioids e-Cigarettes or
nicotine vape, Other tobacco, Marijuana, Other opioids
e-Cigarettes or nicotine vape, Other tobacco, Other
opioids _____

Visit 4

Visit 4 Date

09-15-2025

Have you used opioids or any other substances during the assessment period?☒ No**Visit 5**

Visit 5 Date

09-16-2025

Have you used opioids or any other substances during the assessment period?☒ Yes

Phone

Participant # _____

014

System Usability Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently	☐	☐	☐	☒	☐
2. I found the system unnecessarily complex	☐	☒	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☒	☐
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☐	☒	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☐	☒	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☒	☐	☐	☐

User Engagement Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this app experience.	☐	☒	☐	☐	☐
2. The time I spent in this app just slipped away.	☐	☒	☐	☐	☐
3. I was absorbed in the app task.	☐	☒	☐	☐	☐
4. I felt frustrated while using the app.	☐	☒	☐	☐	☐

5. I found this app to be confusing.	☐	☒	☐	☐	☐
6. Using this app was taxing.	☐	☒	☐	☐	☐
7. The app is attractive.	☐	☐	☒	☐	☐
8. The app is aesthetically appealing.	☐	☐	☒	☐	☐
9. The app appealed to my senses.	☐	☐	☒	☐	☐
10. Using this app was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☒	☐	☐
12. I felt interested in the therapy tasks.	☐	☐	☒	☐	☐

iPAL Application and Intervention Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were videos easy to understand.	☐	☐	☐	☒	☐
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☒	☐

AR

Participant ID _____

014

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Private insurance |
| 3. Which best corresponds to your current work situation? | ☒ Disabled or retired and not looking for work. |

iGROUP Presence Questionnaire

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Disagree |
| 2. I felt that the virtual world surround me. | ☒ Completely Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Disagree |
| 4. I did not feel present in the virtual space. | ☒ Completely Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something form the outside. | ☒ Somewhat Disagree |
| 6. I felt present in the virtual space | ☒ Completely Disagree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely Disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Disagree |
| 11 How real did the virtual world seem to you? | ☒ Very unreal |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat Disagree |
| 13. How real did the virtual world seem to you? | ☒ Very unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat Disagree |

System Usability Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☒	☐	☐	☐	☐
2. I found the system unnecessarily complex.	☐	☐	☐	☐	☒
3. I thought the system was easy to use.	☐	☒	☐	☐	☐
4. I think that I would need the support of a technical person to be able to use this system.	☐	☒	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☒	☐	☐	☐
6. I thought there was too much inconsistency in this system.	☐	☐	☒	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☐	☐	☐	☒	☐
9. I felt very confident using the system.	☐	☐	☒	☐	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☒	☐

User Engagement Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this augmented reality experience	☒	☐	☐	☐	☐
2. The time I spent in augmented reality just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the therapy task.	☒	☐	☐	☐	☐
4. I felt frustrated while using the headset.	☐	☒	☐	☐	☐
5. I found this system to be confusing.	☐	☐	☒	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☒	☐	☐	☐

8. The augmented reality is aesthetically appealing.	☐	☒	☐	☐	☐
9. The augmented environment appealed to my senses.	☐	☒	☐	☐	☐
10. Using this system was worthwhile.	☒	☐	☐	☐	☐
11. The therapy experience was rewarding.	☐	☒	☐	☐	☐
12. I felt interested in the therapy tasks.	☐	☐	☒	☐	☐

iPAL Application and Intervention Evaluation AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☒	☐	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

VR

Participant ID _____

014

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Private insurance |
| 3. Which best corresponds to your current work situation? | ☒ Working part time. |

iGroup Presence Scale

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Disagree |
| 2. I felt that the virtual world surrounded me. | ☒ Completely Agree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Agree |
| 4. I did not feel present in the virtual space. | ☒ Completely Disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Somewhat Agree |
| 6. I felt present in the virtual space. | ☒ Completely Agree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely Disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat Agree |
| 11. How real did the virtual world seem to you? | ☒ Somewhat unreal |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Completely Agree |
| 13. How real did the virtual world seem to you? | ☒ Somewhat unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely Disagree |

System Usability Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☒	☐
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this virtual experience	☒	☐	☐	☐	☐
2. The time I spent in the virtual world just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the therapy task.	☐	☐	☐	☒	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The virtual environment is aesthetically appealing.	☐	☐	☐	☒	☐
9. The virtual environment appealed to my senses.	☐	☐	☐	☒	☐
10. Using this system was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Evaluation VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☐	☒
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☐	☒
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☒	☐

MR

Participant ID _____

014

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Private insurance |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGROUP Presence Questionnaire MR

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Completely Agree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Agree |
| 4. I did not feel present in the virtual space. | ☒ Completely disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Somewhat Agree |
| 6. I felt present in the virtual space. | ☒ Completely Agree |
| 7. How aware were you of the real world surround you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat Agree |
| 11. How real did the virtual world seem to you? | ☒ Somewhat real |
| 12. How much your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat Agree |
| 13. How real did the virtual world seem to you? | ☒ Somewhat real |
| 14. The virtual world seemed more realistic than the real world. | ☒ Completely disagree |

System Usability Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☒	☐
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☒	☐
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this mixed-reality experience.	☒	☐	☐	☐	☐
2. The time I spent in mixed reality just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the therapy task.	☐	☐	☒	☐	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The mixed-reality is aesthetically appealing.	☐	☐	☐	☒	☐
9. The blended environment appealed to my senses.	☐	☐	☐	☒	☐
10. Using this system was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☐	☒
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☐	☒
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

Exit Interview

1. Which option describes your current home?	☒ It is owned or being bought by you or someone in household
2. How do you pay for healthcare?	☒ Private insurance
3. Which best corresponds to your current work situation?	☒ Not working or looking for work.
1. You experienced Cognitive Behavioral Therapy and Heart Rate Variance interventions in virtual, augmented, and mixed realities. How would you describe your experiences with each of those headsets?	believes that all headsets were easy to navigate. MR - felt like there was more of a real image than artificial image and believes that it would be the easiest to navigate. participant states that they actually felt like they were on the beach during the HRV intervention. VR - enjoyed the visuals and enjoyed being able to get a fuller picture of everything. believes that the imaging was too animated and that the headset itself was too bulky. AR - did not enjoy at all, but believes that aspect was nice. states that it was a bit more difficult to navigate initially and visuals were more difficult to see.
2. Do you think these types of technologies have a place in OUD treatment and management? If YES, how? If NO, why not?	does not believe that they would be beneficial during crisis mode in a real life scenario. states that it would work if these headsets were available, but does not think that this is logical or realistic due to how expensive these headsets are. notes that it could help, but there has to be practicality when developing a treatment plan and making these interventions more accessible.
3. How would you describe the iPAL application, in terms of a cravings management tool?	asked the right questions and was not too long or too short. hit the main points that are important.
4. How would you describe the interventions (CBT, HRV)? Were the interventions appropriate for the intensity level and triggers you reported?	CBT - perfect length, not too long or too short; could definitely help in coming to senses and thinking about aspects of life in different ways. notes that the videos personally did not help the participant with the intensity of their cravings, but helped identify triggers for cravings. HRV - very good intervention; helped calm down and get in the right headspace and believes it was a perfect length. both interventions reported to be appropriate for intensity level and triggers reported.
5. Do you think the iPAL app, or similar digital interventions delivered via phone should be integrated into OUD treatment plans? If YES, how and in what capacity or role? if NO, why not?	yes - could definitely assist in identifying triggers and allowing space for self. reports that identifying triggers is one of the hardest things to do for someone in addiction and some people don't know/can't identify why they like getting high, so this could be very beneficial for someone in active addiction.
6. If iPAL or a similar app was available, would you download the app and incorporate it into your OUD treatment and management plan?	yes - could envision self utilizing app 3-4x/week